

Ensemble - I

Agenda

- Ensemble
- Types of Ensemble methods
- Bootstrap Sampling
- Bagging
- Random Forest
- Voting
- Ensembles for Classification and Regression
- Code Walkthrough

Ensemble

Agenda

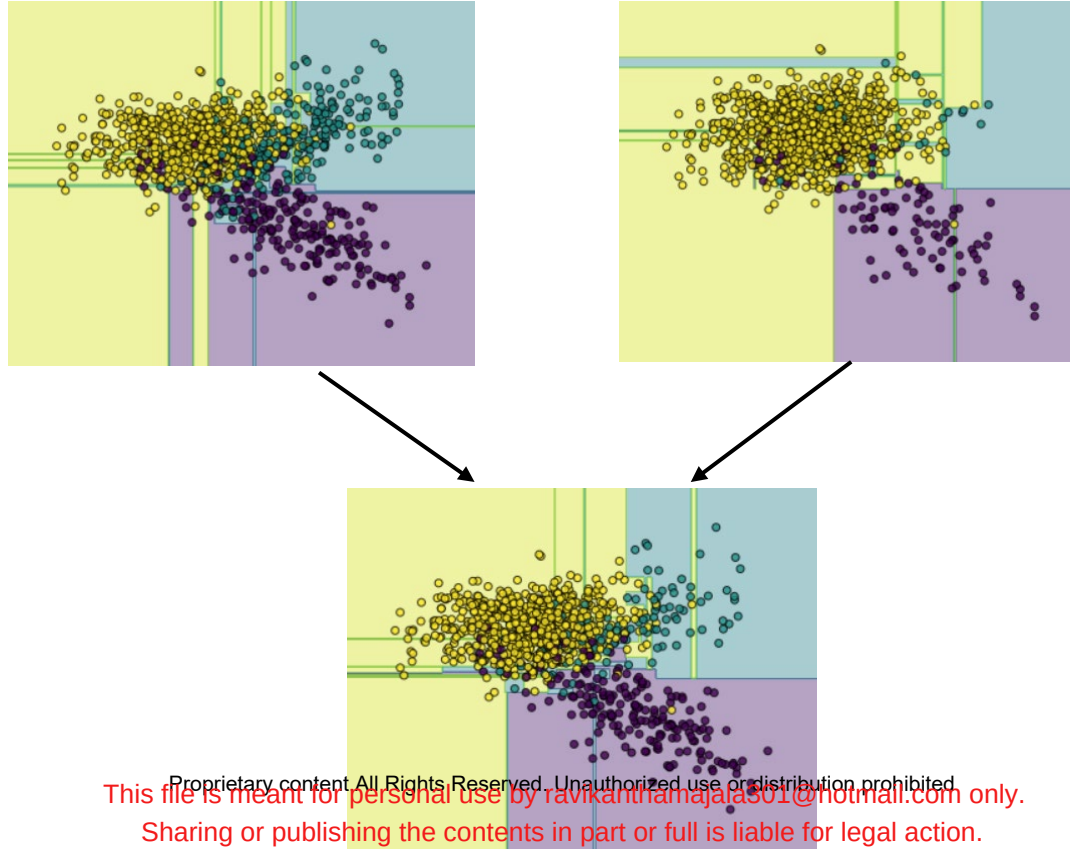
- What is an Ensemble method?
- Bias and Variance Error
- Why Ensemble methods?
- Prerequisites for effective ensemble

What is an Ensemble Method?

Bias and Variance error

Why Ensemble Methods?

Why Ensemble Methods?



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Prerequisites for effective ensemble

Summary

Types of Ensemble methods

Types of Ensemble

- Bagging
- Boosting

Bagging

Boosting

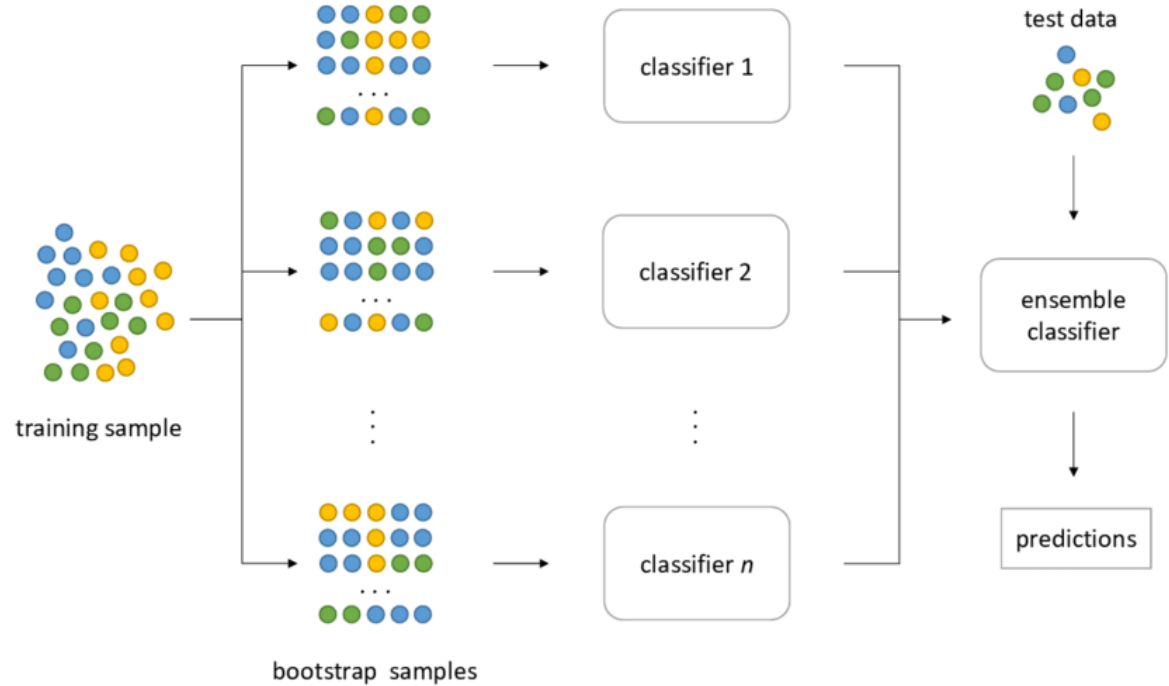
Summary

Bootstrap Sampling

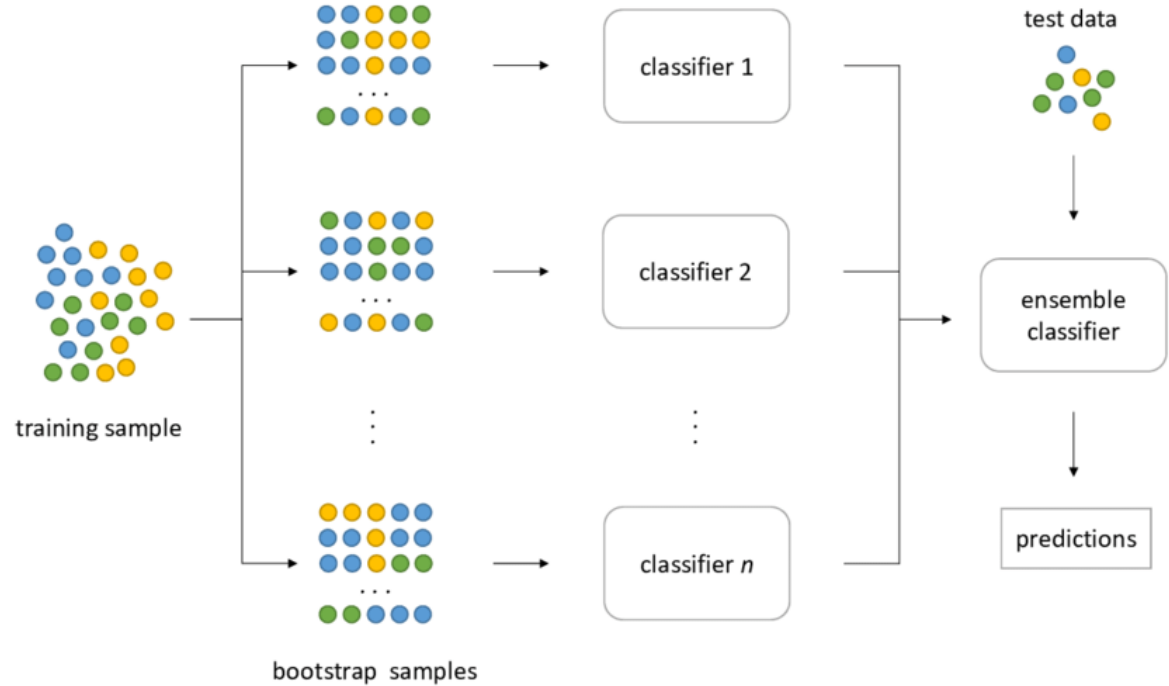
Agenda

- Bootstrap Sampling
- Out of Bag samples

Bootstrap Sampling



Bootstrap Sampling



Bootstrap Sampling

Data	1	2	3	4	5	6	7	8	9	10
BS1	5	5	4	2	7	1	4	8	8	3
BS2	3	1	1	2	8	5	2	10	7	4
BS3	2	8	8	5	4	7	1	3	3	10

Out of Bag samples

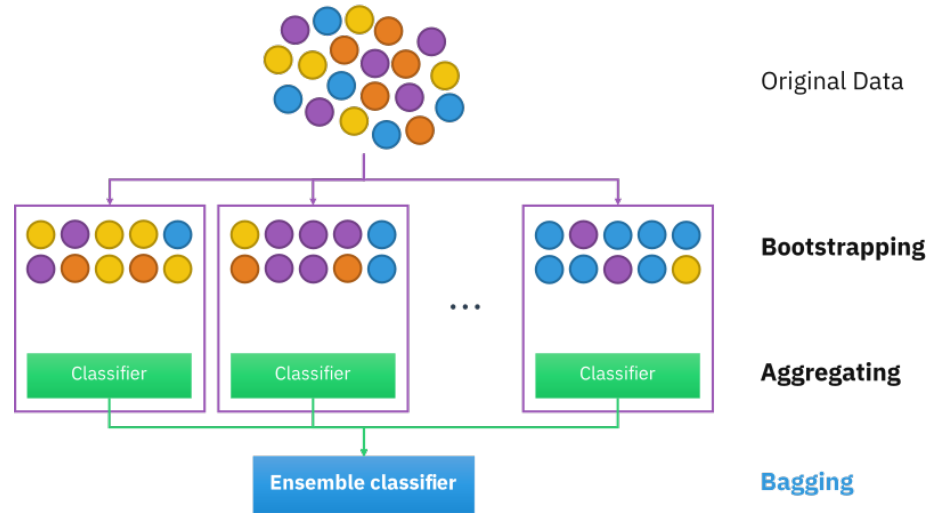
Out of Bag samples

Da ta	1	2	3	4	5	6	7	8	9	10
S1	5	5	4	2	7	1	4	8	8	3
S2	3	1	1	2	8	5	2	10	7	4
S3	2	8	8	5	4	7	1	3	3	10

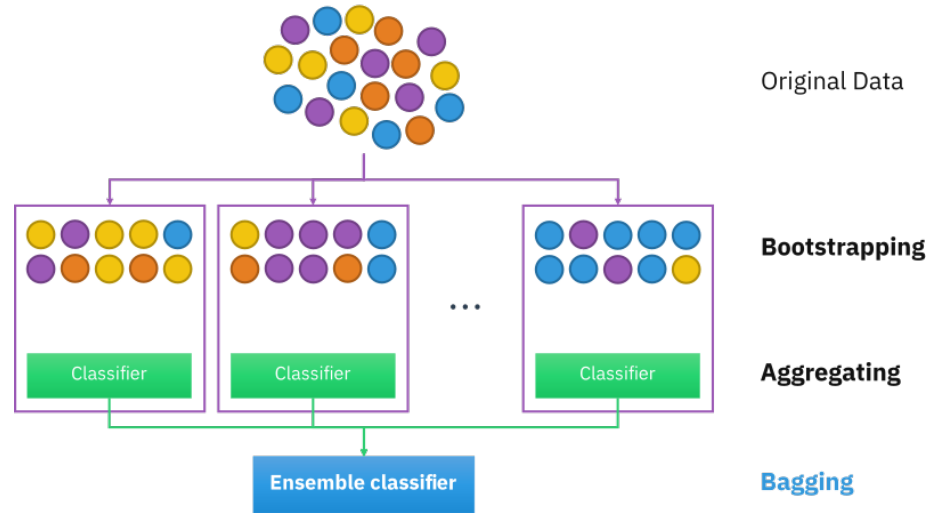
Summary

Bagging

What is Bagging?



What is Bagging?



Summary

Random Forest

What is Random forest model?

What is Random forest model?

Summary

Voting

Agenda

- What is Voting?
- Types of Voting
- Soft Voting
- Hard Voting

What is Voting?

Types of Voting

- Hard Voting
- Soft Voting

Hard Voting

Soft Voting

Summary

Ensembles for Classification and Regression

Classification

Regression

Summary

Code Walkthrough

Summary

Thank you